

GI hormones

Maneerat Chayanupatkul, MD.

Department of Physiology, Faculty of Medicine, Chulalongkorn University





Outline of talk

1. **Essential GI hormones – related disorders**

Gastrin – Zollinger-Ellison syndrome, CCK – abdominal pain, Secretin – pancreatic insufficiency, VIP – VIPoma, Somatostatin – portal hypertension, neuroendocrine tumors

2. **Insulin-related GI hormones**

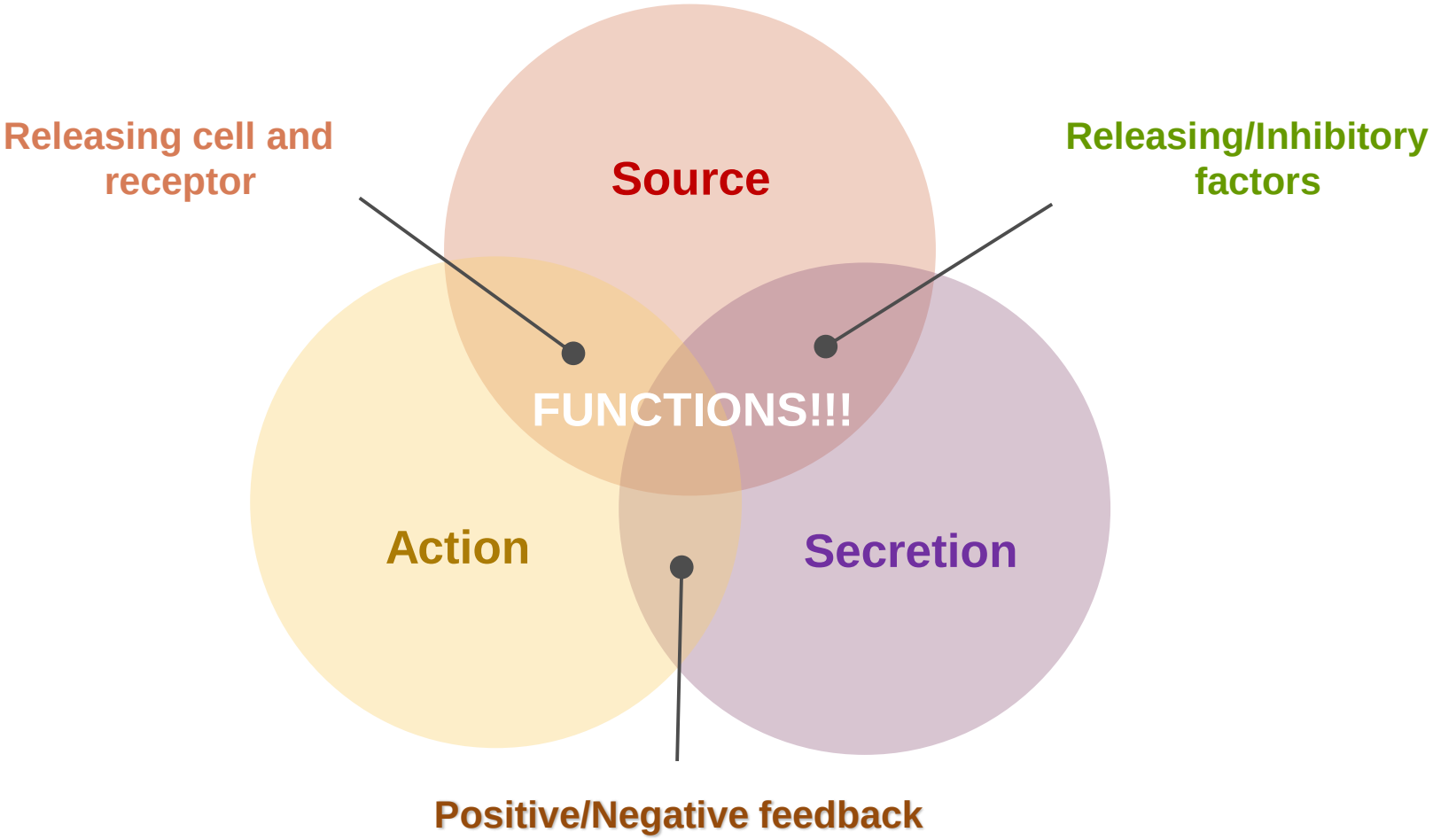
GIP and GLP-1 – diabetes mellitus

3. **Other GI hormones**

Motilin – migrating motor complex, Guanylin – *E coli*



Hormone regulation





Summary of GI hormones

Hormone	Actions	Stimulants	Inhibitors	Clinical application
Gastrin	- Stimulate gastric acid and pepsin secretion - Stimulate of gastric motility	Peptide/amino acids in the stomach, high gastric pH, vagus nerve stimulation, gastric distention	H^+ in the stomach, somatostatin, secretin	Zollinger-Ellison syndrome, *H.pylori* infection (antral type vs. corpus type)
Cholecystokinin	- Induce gall bladder contraction and sphincter of Oddi relaxation - ↑ pancreatic enzyme secretion	Peptides, amino acids and fatty acids in the small intestine	Lack of nutritional stimulation	Reason for biliary pain (worsen with fatty food) CCK-augmented HIDA scan



Summary of GI hormones

Hormone	Actions	Stimulants	Inhibitors	Clinical application
Secretin	↑ $HCO3^-$ secretion by duct cells of pancreas and biliary tract→ ↑secretion of a watery, alkaline pancreatic juice Acts through CFTR	Protein digestive products, fatty food and increased acidity in duodenal content (pH< 4.5-5)	Alkaline juice in duodenum (negative feedback), somatostatin	Secretin stimulation test in ZES and exocrine pancreatic insufficiency Secretin-enhanced MRCP in the evaluation of pancreatic disease



Summary of GI hormones

Hormone	Actions	Stimulants	Inhibitors	Clinical application
VIP	- ↑ secretion of electrolyte and water from small bowel - ↑ intestinal blood flow - Relaxation of intestinal smooth muscle especially sphincters (LES)	Esophageal and gastric distention, vagal stimulation, fatty acid and ethanol in duodenum		VIPoma (watery diarrhea-hypokalemia-achlorhydria, flushing)



Summary of GI hormones

Hormone	Actions	Stimulants	Inhibitors	Clinical application
Somatostatin	- Inhibit all functions of GI tract - Reduce splanchnic blood flow	Gastric acid, dietary components		Somatostatin analogue – used in variceal bleeding, neuroendocrine tumor
GLP-1	- Stimulate insulin release from pancreatic islet - Reduce appetite - Cardioprotective effect	Glucose and fat in duodenum		GLP-1 agonist in diabetes and to induce weight loss in obesity



Summary of GI hormones

Hormone	Actions	Stimulants	Inhibitors	Clinical application
Motilin	- Major regulator of MMC			Erythromycin (binds to motilin receptor) → uses in gastroparesis, pre-endoscopy in GIB
Guanylin	↑ Cl secretion through Cl channel into small intestine → ↑ luminal $HCO3^-$, Na^+ and H2O			Heat stable enterotoxin of *Escherichia coli* activates guanylin receptors → diarrhea



Zollinger-Ellison syndrome (ZES)

- Triad of **gastrinoma**, **hypergastrinemia**, **severe peptic ulcer (PU)** disease
- Gastrinoma most commonly found in duodenal wall followed by pancreas
- Signs and Symptoms (S&S): PU refractory to treatment, multiple ulcers, ulcers in distal part duodenum, jejumun, gastroesophageal reflux disease (GERD), diarrhea, **steatorrhea**, weight loss and vitamin B12 malabsorption



Zollinger-Ellison syndrome (ZES)

- Serum gastrin of >1000 with gastric pH<2 are highly suggestive of ZES
- Secretin stimulation test can be used in equivocal cases
- Normal gastric G cells will be inhibited by secretin but gastrinoma cells will be stimulated by secretin
- Abrupt/high level of gastrin after secretin injection is diagnostic of ZES

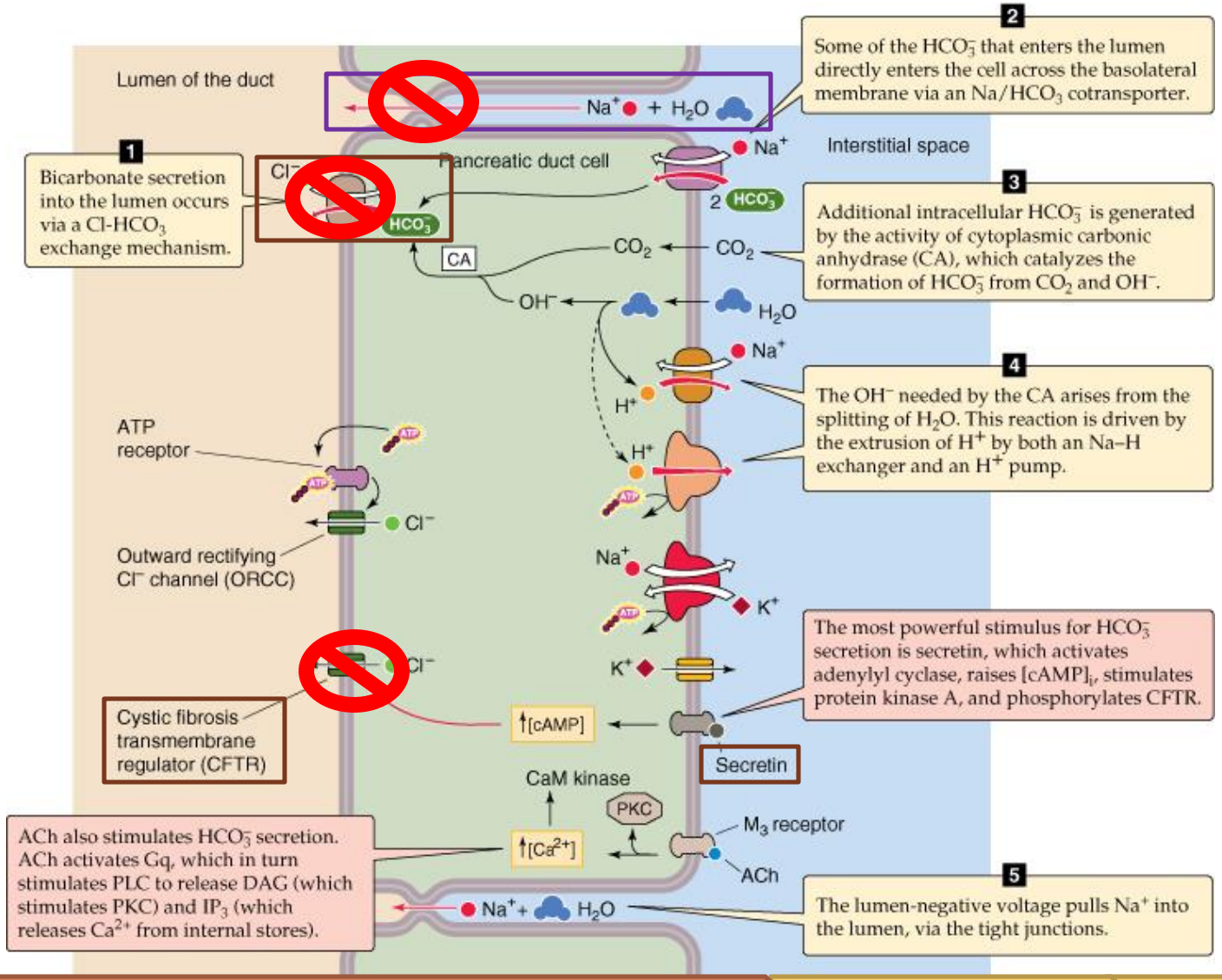




Helicobacter pylori (H.pylori) gastritis

- H.pylori infection can affect gastric acid secretion differently depending on location
- **Antral-predominant** H.pylori gastritis >> cytokines release >> disrupt D cell function >> ↓ somatostatin >> hypergastrinemia >> high acid secretion >> duodenal ulcer
- **Corpus-predominant** H.pylori gastritis >> atrophy in gastric body >> parietal cell loss >> hypochlorhydia (low acid) >> stimulate gastrin >> hypergastrinemia >> ↑ gastric cancer risk from trophic effect from gastrin

Secretin and cystic fibrosis



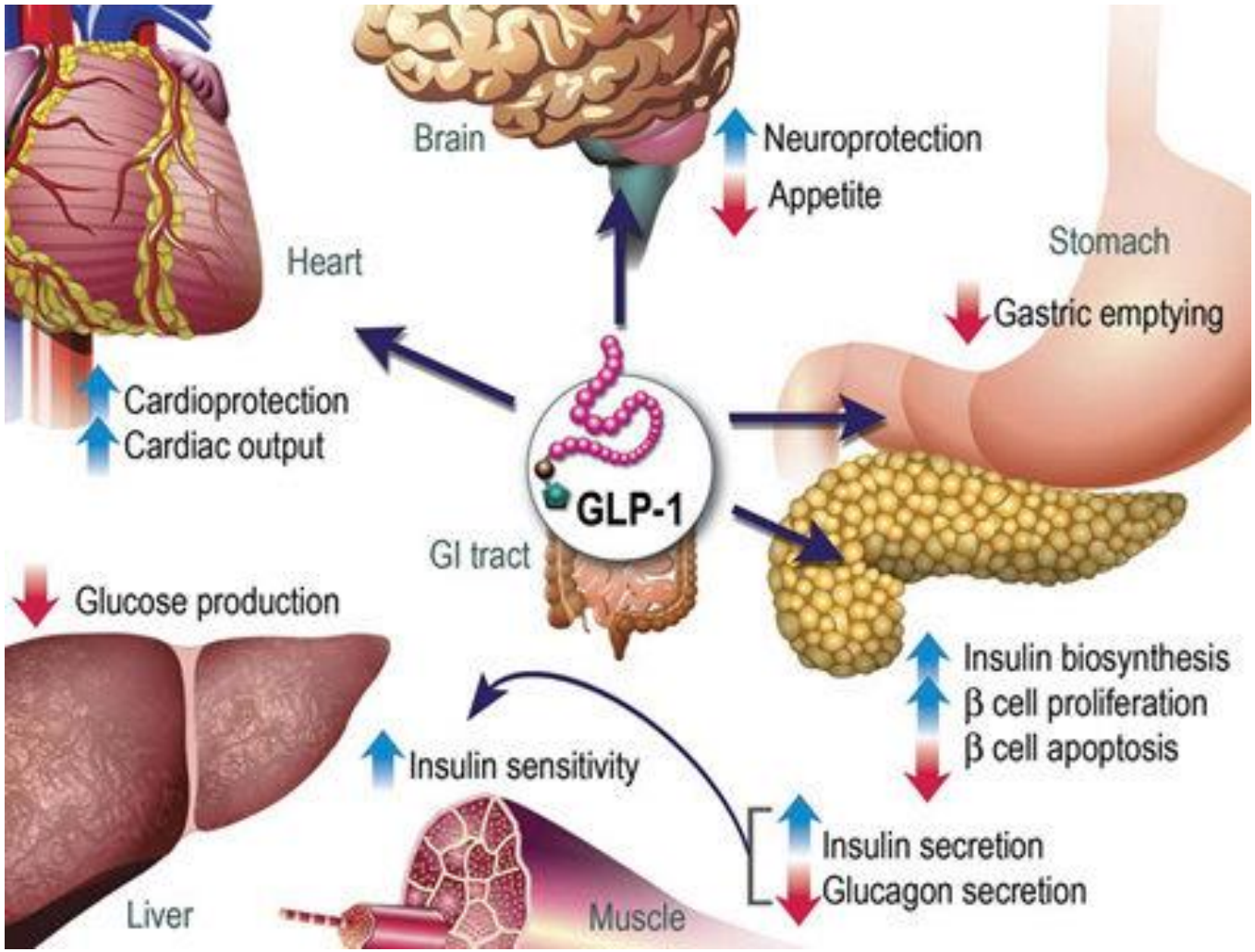


Somatostatin – clinical applications

- Somatostatin analog or octreotide can be used in several conditions
- Treatment of neuroendocrine tumors
- Variceal bleeding (to reduce portal pressure)
- Reduce stool volume in short gut syndrome or other causes of secretory diarrhea
- Localization of NET (Octreotide scan)



GLP-1 – actions



GLP-1 (Glucagon-like peptide 1)

– clinical applications

- **GLP-1 agonist (-natide/-glutide)** and **DPP-IV inhibitor (gliptins)** are treatment options for type 2 DM (↓ post-prandial hyperglycemia, ↑Beta cell mass)
- Negligible risk of hypoglycemia



Motilin – clinical applications

- **Erythromycin** bind to motilin receptor → ↑ GI motility → side effects of cramping and diarrhea/can also be used to increase gastric emptying in patients with gastroparesis or before endoscopy





Guanylin – source and action

- Act via C-type guanylyl cyclase → ↑ cGMP → ↑CFTR → ↑ Cl secretion through Cl channel into small intestine → ↑ luminal HCO_3^-, Na^+ and H_2O
- Heat stable enterotoxin of ***Escherichia coli*** (*E.coli*) has very similar structure to guanylin → activate guanylin receptor in small bowel → diarrhea



Thank your for your attention

